

16.6

Parametric Surfaces

Goals:

1. To find the parametric equations of a surface.
2. To calculate surface area.

Definition:

A vector-valued function of two input variables can describe a surface, S , in space:

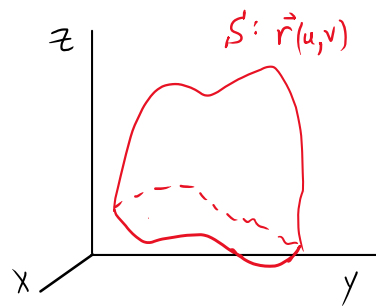
$$\vec{r}(u, v) = x(u, v)\vec{i} + y(u, v)\vec{j} + z(u, v)\vec{k}$$

(u, v) lies in a set, D domain

Note:

$x = x(u, v)$, $y = y(u, v)$, and $z = z(u, v)$ are called the parametric equations of S
and u and v are called parameters

$$\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$$



Example:

Parameterize...

- a) The upper hemisphere of $x^2 + y^2 + z^2 = 1$

$$z = \sqrt{1 - x^2 - y^2}$$

$$f(x, y) = \sqrt{1 - x^2 - y^2}$$

$$x = u, y = v, z = \sqrt{1 - x^2 - y^2}$$

$$\vec{r}(t) = u\vec{i} + v\vec{j} + \sqrt{1 - x^2 - y^2}\vec{k}$$

- b) A sphere of radius a .

$$y = f(x)$$

$$t = x, y = f(t)$$

$$\vec{r}(t) = \langle t, f(t) \rangle$$

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parametric eqns.

$$x^2 + y^2 + z^2 = a^2$$

$$\begin{cases} x = a \sin \phi \cos \theta \\ y = a \sin \phi \sin \theta \\ z = a \cos \phi \end{cases}$$

borrow from spherical coord. w/ $\rho = a$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \phi \leq \pi$$

c) $4x^2 + y^2 = 16$

elliptic cylinder!

$$\frac{4x^2}{16} + \frac{y^2}{16} = 1$$

$$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$$

$$\left. \begin{aligned} x &= 2 \cos u \\ y &= 4 \sin u \\ z &= v \end{aligned} \right\} \begin{aligned} 0 &\leq u \leq 2\pi \\ v &\in \mathbb{R} \text{ (set of real numbers)} \end{aligned}$$

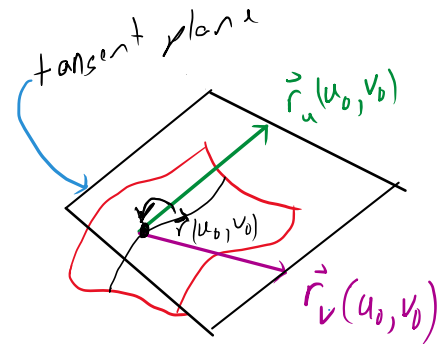
$$\vec{r}(u, v) = \langle 2 \cos u, 4 \sin u, v \rangle, \quad 0 \leq u \leq 2\pi, \quad v \in \mathbb{R}$$

Definition:

Let a surface, S, be given by $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$:

$$\mathbf{r}_u(u, v) = \frac{\partial x}{\partial u} \vec{i} + \frac{\partial y}{\partial u} \vec{j} + \frac{\partial z}{\partial u} \vec{k}$$

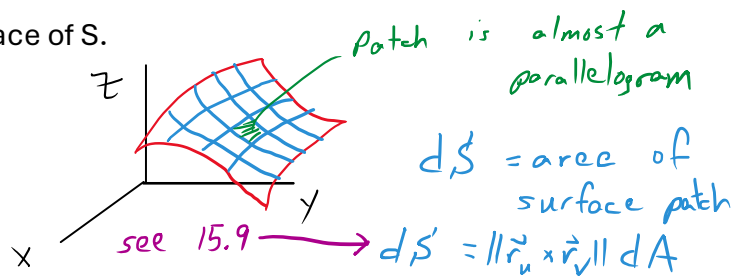
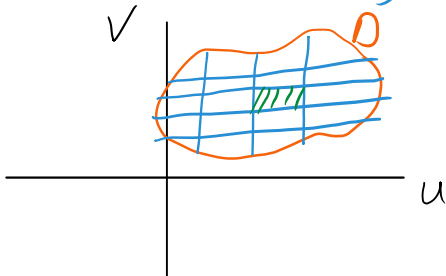
$$\mathbf{r}_v(u, v) = \frac{\partial x}{\partial v} \vec{i} + \frac{\partial y}{\partial v} \vec{j} + \frac{\partial z}{\partial v} \vec{k}$$



Notes:

- Both $\mathbf{r}_u(u, v)$ and $\mathbf{r}_v(u, v)$ are tangent vectors to S at $\mathbf{r}(u, v)$.
- As long as $\vec{r}_u(u_0, v_0) \times \vec{r}_v(u_0, v_0) \neq \vec{0}$, S has a tangent plane at $\mathbf{r}_u(u_0, v_0)$.
- S is called smooth if $\vec{r}_u \times \vec{r}_v \neq \vec{0}$ for all (u, v) in the domain, D, of $\mathbf{r}(u, v)$.
- As long as S is smooth, $\mathbf{r}(u, v)$ maps small rectangles in D to small

"parallelograms" on the surface of S.



- As long as S is smooth, the surface area of S is

$$A(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| dA$$

- If S is given by $z = f(x, y)$, the S can be expressed parametrically as

$$\langle x, y, f(x, y) \rangle$$

and thus $\|\mathbf{r}_u \times \mathbf{r}_v\| =$

$$\sqrt{1 + f_x^2 + f_y^2}$$

since

$$\begin{aligned} \mathbf{r}_x &= \langle 1, 0, f_x \rangle \\ \mathbf{r}_y &= \langle 0, 1, f_y \rangle \end{aligned}$$

$$\begin{aligned} &= -f_x \vec{i} - f_y \vec{j} + 1 \vec{k} \\ &\| -f_x \vec{i} - f_y \vec{j} + 1 \vec{k} \| = \sqrt{f_x^2 + f_y^2 + 1} \end{aligned}$$

Surface Area of a Smooth Surface Given by $z = f(x, y)$:

$$A(S) = \iint_R \sqrt{1 + f_x^2 + f_y^2} \, dA, \text{ assuming domain of } f \text{ is } R.$$

Example:

Find the area of $f(x, y) = xy$ with domain $R = \{(x, y) | x^2 + y^2 \leq 1\}$

$$A(S) = \iint_R \sqrt{1 + f_x^2 + f_y^2} \, dA$$

$$f_x = y, \quad f_y = x$$

$$A(S) = \iint_{x^2 + y^2 \leq 1} \sqrt{1 + y^2 + x^2} \, dA$$

$$= \int_0^{2\pi} \int_0^1 \sqrt{1 + r^2} \, r \, dr \, d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^1 \sqrt{1 + r^2} \, 2r \, dr$$

$$= \frac{1}{2} \cdot 2\pi \cdot \frac{2}{3} \left[(1 + r^2)^{\frac{3}{2}} \right]_0^1 = \frac{2\pi}{3} \left[2^{\frac{3}{2}} - 1 \right] = \frac{2\pi}{3} [2\sqrt{2} - 1] \text{ units}^2$$

surface area of S (given by $z = f(x, y)$) restricted to inputs on unit disk

Summary

(1) $\vec{r}(u, v)$ can represent a surface in space

(2) Surface area of S with domain R : $\iint_S dS = \iint_R \sqrt{1 + f_x^2 + f_y^2} \, dA$
 given by $z = f(x, y)$ (x, y) comes from R

